

CROSSSAFE · REQUIREMENTS

CrossSafe requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	R9 met on paper for 76 to 114.3 mm poles; R10 and R15 figures updated (cost \$364.00 per assembly)
0.7	2026-10-02	Amish Chadha	R10 restated on 2026-10-02: existing-pole kit 76 to 114 mm until a slip-torque test, panel socket for free-topped poles of about 114 mm; no status changed
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Status from CRS-CAL-001 v0.3 for the constructable design (CRS-DDR-003); R15 now not met on paper
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from CRS-CAL-001; R15 redefined to the existing-pole assembly (CRS-DDR-001 D1); R1, R9 and R11 updated for D3, D4 and D8
0.2	2026-09-25	Amish Chadha	

First measurable requirements for TRL 2, with status against the concept estimates

0.1	2026-09-25	Amish Chadha	Initial scaffold
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CrossSafe requirements

These requirements are checked by calculation in CRS-CAL-001 at TRL 3. Targets are still proposals, not yet validated with users or a road authority, and will be revised after co-design (see CRS-PRB-001). Where a local rule is stricter, the local rule wins. The design choices behind them were accepted by Amish on 2026-09-25 (CRS-DDR-001 and CRS-DDR-002).

The **design case** is a marked crossing on a two-lane road 7 m kerb to kerb, one CrossSafe assembly on each side, 300 activations of 20 s per day, 2.5 peak sun hours per day in the worst month, and an ambient range of -20 to +50 °C.

Table 1. Requirements and status at TRL 3 (CRS-CAL-001 v0.4, Table 5).

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
R1	Warning light geometry and pattern	Two amber rectangular indications per face, each at least 127 x 51 mm (5 x 2 in); the FHWA IA-21 wig-wag sequence at 75 flashing sequences per minute; pattern set in firmware for other jurisdictions; an optional pedestrian pilot light where allowed	Design review against the pilot jurisdiction's rules	Met by design (140 x 62 mm lenses)
R2	Visibility	Indications distinguishable by a driver at 100 m or more in direct sun (IA-21 cites SAE J595 Class 1 yellow peak intensity) and not glaring at night (automatic dimming)	Photometric calculation from LED data; later field check	Not verifiable at TRL 3 (screening: about 3,939 cd per head)
R3	Activation	Push button and passive presence detection; both sides of the road flash within 0.5 s of a trigger on either side	Timing budget; later bench test	Met on paper (48 ms; 244 ms with two retries)
R4	Flash duration	Set per site from crossing length / 0.9 m/s + 5 s, adjustable 10 to 40 s; a new trigger while flashing extends the time	Firmware review	Met by design (12.8 s for 7 m)
R5	Detection quality	Passive detection triggers for a person waiting 1 s or more in the waiting zone; false activations 5 % or fewer of all activations; missed waiting pedestrians 2 % or fewer	Site trial with manual counts	Not verifiable at TRL 3
R6	Autonomy without sun	5 days or more of design-case use from a full battery	Energy calculation	Met on paper (7.7 days; 5.4 at -20 °C)
R7	Energy balance	Energy neutral at 2.5 peak sun hours; recover from 20 % to full in 3 days or fewer at 5 peak sun hours	Solar yield calculation	Met on paper (break-even 1.05 h; recovery 2.1 days)
R8	Environment	Enclosures and connectors IP65 or better; operate -20 to +50 °C ambient; battery charges only at 0 to 45 °C cell temperature	Datasheets and heat balance; later spray and thermal tests	Met on paper with the sun shield (53.9 °C inside at 50 °C ambient when dusty; charging to 41.1 °C ambient)
R9	Structure	Post and clamps carry the sign and panel in a 40 m/s gust with stress 60 % or less of yield; new posts are 114.3 x 3.6 mm (CRS-DDR-001 D3); the sign cannot rotate on its pole (anti-rotation bolt on new posts, keyed saddles on existing poles, CRS-DDR-002)	Wind load calculation	Met on paper for poles of 76 to 114.3 mm: post 45 % of yield and bolt 12 % of bearing; keyed saddles resist 191 N·m on a 76 mm pole against 175 N·m, at assumed friction and band tension
R10	Mounting	Fits the project's own post or an existing 76 to 114 mm pole with band clamps (60 to 75 mm poles only after a slip-torque test confirms the keyed saddle's grip, with a third saddle and band at the sign); the prototype's panel socket fits free-topped poles of about 114 mm; sign bottom at 2.1 m or	Design review; later timed trial	Not verifiable at TRL 3 (fit met on free-topped poles of about 114 mm; install estimate 120 min, at the limit)

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
		more (or local rule); two-person crew installs one assembly in 2 h or less		
R11	Accessibility	Push button 0.9 to 1.1 m above the walking surface, operable with 22 N or less, with an audible and tactile acknowledgement and a pedestrian-facing pilot light	Design review with disability groups	Met by design (button at 1.05 m), to be checked in co-design
R12	Fail-safe behavior	Never flashes continuously; a fault (low battery, open LED, lost radio link) is shown on a maintenance indicator and logged; with the link lost, each side still flashes on a local trigger	Firmware review; fault injection later	Met by design
R13	Privacy	No images or audio are recorded or leave the device; only activation counts, faults and battery state are logged or sent	Design review	Met by design
R14	Theft and vandal resistance	Battery, charger and controller 2.8 m or more above the walking surface; security fasteners; no exposed cables below 2.5 m	Design review	Met by design (enclosure bottom 2.85 m; sensing head 2.515 m and up)
R15	Affordable	Parts for one assembly on an existing pole \$350 or less (the prototype budget_usd covers, CRS-DDR-001 D1); parts for a two-sided crossing on existing poles \$700 or less. Crossings that need new posts are costed and reported, not held to this target	Priced BOM (bom/ bom.csv)	Over the value-engineering target by \$14.00 per assembly (\$364.00 against \$350) and \$28.00 per crossing (\$728.00 against \$700); new posts \$437.00 and \$874.00

■ Requirements at risk or not verifiable

One requirement is over its value-engineering target on paper (R15, cost) and three need evidence that a paper study cannot give.

- **R15 over the value-engineering target after the design for construction (CRS-DDR-003).** The saddles, mounting plate, panel bracket, radar arm, battery strap and fixings that make every part buildable and fixed add \$28.00, and the bezels and visors on the four LED heads add \$4.00: the estimated cost of one assembly on an existing pole is \$364.00 against the \$350 target (\$14.00 over), and of a crossing \$728.00 against \$700 (\$28.00 over). The target is a hypothetical control target, not a limit; cost drivers and savings worth trying are in the Value engineering section of the design decisions register.
- **R9 met on paper on poles of 76 mm and larger.** The 114.3 x 3.6 mm post meets the stress target at 45 % of yield, and a 1,800 mm embedment in a 500 mm footing is a screening size. On new posts the M10 through-bolt stops the sign rotating with a wide margin. On existing poles the keyed saddles hold on poles of about 70 mm and larger at an assumed grip friction of 0.4, so R9 is met on paper for the 76 to 114.3 mm range the existing-pole kit is now limited to (decided 2026-10-02, CRS-DDR-002, N1). The friction and band tension are assumptions, and the 191 N·m resistance on a 76 mm pole is only 9 % above the 175 N·m load. A slip-torque

test on the keyed saddle must confirm the grip before the kit is used below 76 mm, when a third saddle and band would bring 60 to 75 mm poles back. On existing poles the host pole must carry about 2.5 kN·m more at the sidewalk, which the owner must confirm.

- **R8 met on paper after the sun shield (CRS-DDR-002).** Unshielded, a dusty enclosure ran 14.7 K above ambient and reached 64.7 °C at 50 °C ambient; with the ventilated shield the rise is 3.9 K and the interior 53.9 °C, below the typical 60 °C discharge limit. The shield factor is borrowed, not measured.
- **R2, R5 and R10 are not verifiable at TRL 3.** R2 and R5 decide whether drivers respond and whether passive detection is trusted.
- **R6, not met at TRL 2, is now met on paper** after the PIR gating (D2).

■ Assumptions

The full list is in CRS-CAL-001, Table 1. The main ones are:

- LED heads draw about 6 W each when lit in daytime and are lit 25 % of each IA-21 sequence; LED drivers are about 91 % efficient.
- Standby about 187 mW: radio in continuous receive and controller about 23 mW, 12 V quiescent about 13 mW, charger about 51 mW, radar 0.4 W on 25 % of the day under PIR gating.
- Solar system efficiency about 76 % from panel rating to energy stored (panel heat and dust 0.85, MPPT 0.94, cell charge 0.95).
- Walking speed of 0.9 m/s allows for children and older people; the US MUTCD basis is faster, and local rules decide.
- Gust wind speed of 40 m/s and a drag coefficient of 1.2 for every part; S235 steel for the post; band tension 1,000 N; keyed saddle grip friction 0.4; a ventilated shield passes 25 % of the solar gain.